

Independent Events – Multiplication Rule

- Two events are independent if they _____.
- The occurrence of one _____ on the occurrence of the other.
 - Toss a coin twice
 - The gender of each child born to the same mother
 - Trials with replacement
- If A and B are independent events, then the probability that both A and B occur is found using the *multiplication rule*:

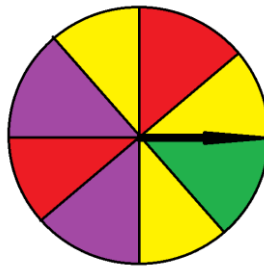
Examples) Find the probabilities.

a) I flip a coin and then roll a dice. What is the probability I flip a head and then roll a 5?

b) What is the probability that Mrs. G will have 4 girls in a row?

c) I have a standard deck of 52 cards. What is the probability I pull a red card, replace it and shuffle, and then pull a black card?

d) You spin the following spinner 5 times. What is the probability that you land on yellow all 5 times?

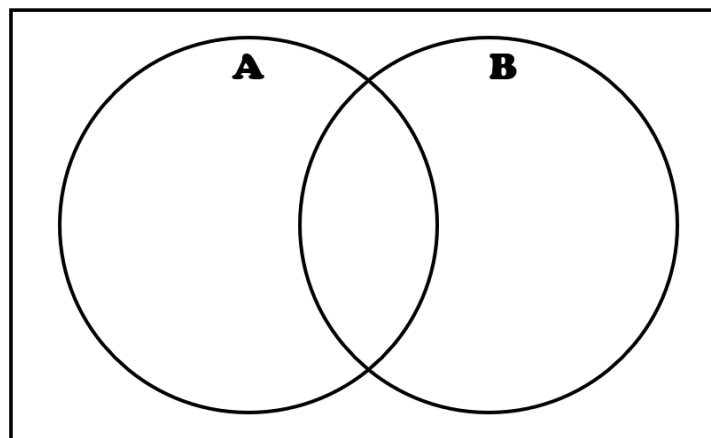


Dependent Events – Multiplication Rule

- Events are dependent if they _____.
- The occurrence of one _____ the occurrence of the other.
 - Trials without replacement
- If A and B are dependent events, the probability that both A and B occur is found by
- Where $P(B|A)$ is the probability of _____
- The *conditional probability* of an event is the probability that one event will happen, if it is known that another event *has* happened.
- We can rearrange the multiplication rule a little and get a formula for conditional probability:

Why does this formula make sense?

- When you know something has happened, that changes the sample space – specifically, it shrinks to the given event.
- If we know A has already occurred, that becomes your “sample space”
- Then, for B to occur, it has to occur with A, which is the intersection of A and B



$$P(B|A) = \text{_____}$$

Example 1) Find the probabilities.

a) I have a standard deck of 52 cards. What is the probability of pulling a spade and then a 5?

b) I have a standard deck of 52 cards. What is the probability of pulling a jack and then another jack?

c) In your statistics class, you are interested in exploring the taste preference of your classmates. You have two tastes, sweet and salty. 25 students were asked if they prefer both, only sweet, only salty, or neither. The following two-way table shows the data.

	Yes Sweet	No Sweet	Total
Yes Salty	6	7	13
No Salty	11	1	12
Total	17	8	25

What is the probability you select someone who likes salty food and then select another person who likes salty food?

What is the probability you select someone who does not like sweet food and then select a person who likes sweet food?

Example 2)

<u>Hair Color</u>	<u>Eye Color</u>				Total
	Brown	Blue	Hazel	Green	
Black	36	9	5	2	52
Brown	66	34	29	14	143
Red	16	7	7	7	37
Blonde	4	64	5	8	81
Total	122	114	46	31	313

a) If we select a woman and find she has brown hair, what is the probability that she will also have hazel eyes?

b) If we select a woman and find she has brown eyes, what is the probability that she also has black hair?

Showing Independence

- If we draw two cards with replacement, we know that those events are _____.
- If we draw two cards without replacement, we know that those events are _____.

However, in the real world, it can be difficult to determine if two events are independent or dependent. We do have a way to prove independence, however, using either of the following formulas:

Example 2) Below is a two-way table that shows produce and how many packages are purchased based on whether it is fresh or frozen.

	Fresh	Frozen	Total
Blueberries	20	36	56
Pineapple	21	39	60
Avocado	9	15	24
Total	50	90	140

a) Are “Blueberries” and “Fresh” independent?

b) Are “Avocados” and “Frozen” independent?